University
Mohammed VI
Polytechnic

College of
Computing

Lab 6: SQL DDL and DML

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering
Session: Fall 2026

Team Information

Team Name	Team 1
Member 1	Nizar El Hazziti
Member 2	Riyad Habibi
Member 3	Chaimae Iaa
Member 4	Mohamed Jbari
Repository Link	`https://github.com/Niz-arch/Data-Base-Project`

Team roles

Riyad Habibi	Part 1 (DDL)
Chaimae Iaa	Part 2 (data population and updates)
All four team members	Part 3 (SQL queries)
Nizar El Hazziti	Part 4 (open question)
Mohamed Jbari	Report

Implementation difficulties

The order of the inserts mattered because of the foreign keys. We inserted people before their subtype records, equipment models before their units, and stocks before recording consumption. Each consumption record references the lab and consumable together so that it points to the right stock.

For the equipment counts, we used a left join and counted serial numbers so that an empty lab would get a count of zero. Query 14 returns no rows with our sample data because no lab has more than three units.

Assumptions

We used a small sample dataset and November 2026 for the reservation date query. A person with no end date is treated as active. Unused consumables and projects without funding still appear in the totals, with `NULL` when there are no amounts to sum.

Part 4

CE7. Before a person uses an equipment unit, the database must check that they hold every certification required by its model and that each is valid. Foreign keys only check that referenced records exist.

RE1. When a reservation is created or its maker changes, the database must check that the maker is absent from `Administrative`. The foreign key to `Person` only checks that the maker exists.

PR3–PR4. When a project is set up or its participants change, the database must check that it has exactly one principal investigator and that this person is an academic employee. The primary key of `Participates` only guarantees one role per person per project.